Muhammad Awais

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SUMMARY

Results-driven AI/ML Engineer with hands-on experience delivering production-grade **Generative AI**, **NLP**, and **Deep Learning** solutions. Proven ability to reduce inference latency by 40%, achieve 92%+ mAP, and deploy scalable AI systems serving 1,000+ daily users. Proficient in building end-to-end pipelines — from data collection and model training to deployment and MLOps — using **Python, PyTorch, TensorFlow, FastAPI**, and **Scikit-learn**, with a consistent focus on measurable business impact.

PROFESSIONAL EXPERIENCE

AI Engineer — May '25 — Present
BioICAW Technologies — Rawalpindi, Pakistan

- **Led** end-to-end AI engineering projects from data collection and preprocessing to model deployment and production maintenance, ensuring scalable and reliable AI solutions
- Developed and deployed **computer vision** and **NLP** applications achieving sub-50ms inference latency for real-time industrial use cases
- Built production-ready **ML pipelines** with **FastAPI** and **Docker** containerization, implementing **CI/CD** workflows for automated deployment
- Designed data collection frameworks for training datasets including web scraping, API integration, and automated annotation pipelines
- Performed **model optimization** through quantization and **GPU acceleration** using **CUDA**, reducing inference time by 40%
- Implemented **MLOps** practices with **MLflow** for experiment tracking, model versioning, and automated retraining pipelines
- Conducted A/B testing and performance evaluation of deployed models with continuous improvement feedback loops

Data Science Fellow — May '24 — Aug '24
Bytewise Limited — Remote

- Built classification and regression models achieving 90%+ accuracy using **Scikit-learn** and **XGBoost**
- Developed customer churn prediction system with advanced feature engineering reducing churn by 25%
- Conducted comprehensive data analysis and created interactive dashboards for actionable business insights

EDUCATION

BS in Computer Science — Jul '25
Pir Mehr Ali Shah Arid Agriculture University
Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision

PROJECTS

AI Outbound Voice Calling Agent — Real-Time Conversational AI System

- Architected production-grade AI calling agent integrating **OpenAI Realtime API** for live STT and LLM-driven conversation with sub-300ms end-to-end latency
- Engineered hybrid TTS pipeline using **ElevenLabs** voice cloning with WebSocket streaming and HTTP one-shot fallback for accurate email spelling across all voice clones
- Built pre-warm/standby WebSocket connection pooling, eliminating agent silence between turns and reducing response initiation to near-zero latency
- Integrated **Twilio**, **MongoDB**, and **Google Calendar API** for call orchestration, lead management, and automated meeting scheduling; multilingual support (English/Urdu)

KYC Verification Platform — End-to-End Identity & Compliance System

- Built a production-grade **KYC API** using **FastAPI** supporting document verification (CNIC, Passport, Driving License), liveness detection, and 1:1 face matching with sub-50ms inference latency
- Integrated **AML (Anti-Money Laundering)** screening pipeline for automatic sanctions and PEP (Politically Exposed Persons) checks triggered on every document scan, upgrading ACCEPT to REVIEW on hits

- Implemented **Stripe Payment** subscription system with webhook-driven lifecycle management (pending → paid → failed) and tiered usage quotas (Basic/Standard/Premium) tracked per user via **MongoDB**
- Developed anti-spoofing liveness engine processing multi-frame image sequences and video, detecting **FACE_LIVENESS_ERR** and **RECAPTURED_FACE** attacks with confidence-based decision logic
- Built document validation module for Ejari and DEWA PDFs with automated extraction and rule-based verification; deployed with **Docker** and subscription-gated access control

Real-Time Helmet Detection System Using YOLOv8

- Developed **YOLOv8** model trained on 3,500+ annotated images achieving 92.3% mAP; deployed FastAPI inference API with <45ms latency for industrial safety monitoring
- Implemented GPU-accelerated pipeline with **CUDA** optimization reducing inference time by 40%; integrated with live camera feeds at 30+ FPS

AI Medical Assistant — RAG-Powered Diagnostic Support System

- Built **RAG** system processing 10,000+ medical encyclopedia entries with **FAISS**; achieved 93% context relevance using Hugging Face embeddings
- Integrated Groq API LLaMA-70B and GPT models for context-aware retrieval; developed **Streamlit** interface serving 1,000+ daily queries

OratoAI — AI-Powered Public Speaking Coach (FYP)

- Architected a multi-modal **LangGraph agentic AI** system that orchestrates specialized agents across audio, visual, and content analysis pipelines to deliver holistic public speaking feedback
- Built dedicated **voice analysis agent** evaluating Volume, Clarity, Pace, Tone, Consistency, Words Per Minute, and Filler Word detection using speech processing models
- Developed **facial & video analysis agent** leveraging computer vision for Lighting, Stability, Face Presence, Posture, Eye Contact, and Gesture scoring across extracted video frames
- Implemented **NLP content agent** assessing speech Clarity, Structure, Engagement, and Persuasion with automated word count, sentence count, and readability metrics
- Delivered an overall 0–10 composite score with actionable recommendations via a full-stack dashboard (Dashboard, Analyse Speech, History, Profile) built with **FastAPI** backend and modern frontend

SKILLS

Languages: Python, SQL
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, FastAPI, LangChain, FAISS, Streamlit
Computer Vision & Audio AI: YOLOv8, VGG16, OpenCV, Whisper, ElevenLabs
AI/ML Tools: MLflow, Hugging Face, OpenAI API, Groq API
APIs & Integrations: RESTful APIs, Twilio, Stripe Payment, Google Calendar API, MongoDB
DevOps & MLOps: Docker, CI/CD, CUDA, Git, MLflow